Permutations & Combinations

Mind map

Counting principles
- And: multiply; or: add
- At least: complement
- Inclusion-exclusion

Numbers
- Repetition or not
- Divisibility: last digits / sum
- Palindromes, increasing digits

Rank of word
- Alphabetical order
- Smaller letters $\times$ (left)!
- Add 1; alike: divide

Factorial and ${}^nC_r$
- $E_p(n!)$, zeroes $= E_5$
- ${}^nC_r = {}^nC_{n-r}$, Pascal
- Greatest at $r = \frac{n}{2}$

Selection and geometry
- Committee cases
- Lines, triangles, diagonals
- Rectangles, squares, chord points

Gap and string
- Together: block
- Apart: gaps
- Alternate: $2\,n!\,n!$

Station method
- Row: ${}^{n-r+1}C_r$
- Circle: $\frac{n}{n-r}\,{}^{n-r}C_r$

Groups
- $\frac{n!}{p!q!r!}$
- Equal: divide by $k!$
- Each gets one: $2^n - 2$

Alike objects
- $\frac{n!}{p!q!r!}$
- Some at a time: patterns
- Selections $\prod(p+1) - 1$

Circular
- $(n-1)!$
- Garland: divide by 2
- Gap: $(m-1)!\,{}^mP_r$

Divisors
- $d = \prod(a+1)$, $\sigma(N)$
- Two factors, coprime 2^{k-1}
- Product $N^{d/2}$

Grid and derangement
- Paths ${}^{a+b}C_a$
- D_n table
- Place value sum

Distribution of alike
- Beggars ${}^{n+p-1}C_{p-1}$
- Minimum first
- Dummy variable for $\leq$

Coefficient method
- Factor per variable
- Coefficient of x^n

Residues and colouring
- Classes mod m
- Line $k(k-1)^{n-1}$

Partitions, Bell, Stirling
- $P(n,k)$ recursion
- $S(n,k)$, B_n
- Fibonacci, $\varphi(N)$